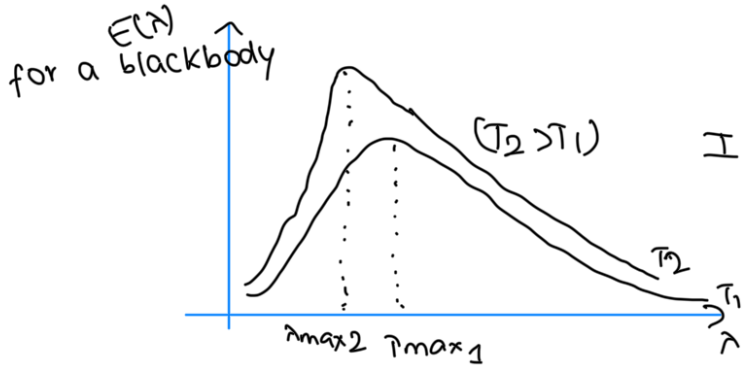
spectral radiance : Power per unit area per unit wavelength



Intensity $I = \int_0^\infty E(\lambda) d\lambda$

(as $T\uparrow$, $\lambda_{max}\downarrow$, $I\uparrow$)

Stefan Boltzman law: $I = \sigma T^4$ $\sigma = 5.67 \times 10^{-8}\ W m^{-2} K^{-4}$

Wein's displacement law: $\lambda_{max} \cdot T = b$ $b = 2898\ \mu m K$

Rayleigh Jean's law: $R(\lambda T) = \frac{8\pi kT}{\lambda^4}$ as $\lambda \to 0$, $R(\lambda) \to \infty$

Planck's law: $E \propto \nu \Rightarrow E = h\nu^{x4}$ $h = 6.55 \times 10^{-34}\ Js$

He changed $R(\lambda, T) = \frac{8\pi ch}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

by curve fitting

Postulates:

→ $\int_{-\infty}^{\infty} \psi \psi^* d\tau = 1$ Normalization

→ Observable → operator $\hat{A}$ $\hat{p}, \hat{E}$

→ $\hat{A}\psi = a\psi$

→ $\langle A \rangle = \int_{-\infty}^{\infty} \psi^* A \psi \, d\tau$

→ $\hat{H}\psi = \hat{E}\psi$

$\Psi = a\psi_1 + b\psi_2 + \dots$

$\Delta x \cdot \Delta p \geq \hbar/2$

$\Delta E \cdot \Delta t \geq \hbar/2$

$\lambda = \frac{h}{p}$

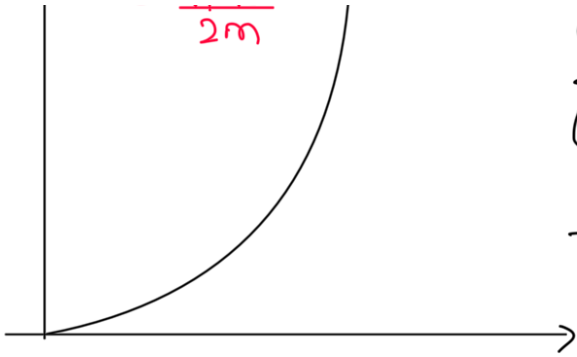
$E = h\nu$

Free electron: $\psi(x) = A e^{i(kx - \omega t)}$

$p = \frac{h}{\lambda} = \hbar k$

$E = h\nu = \hbar\omega$

E ↑ $E = \hbar^2 k^2$



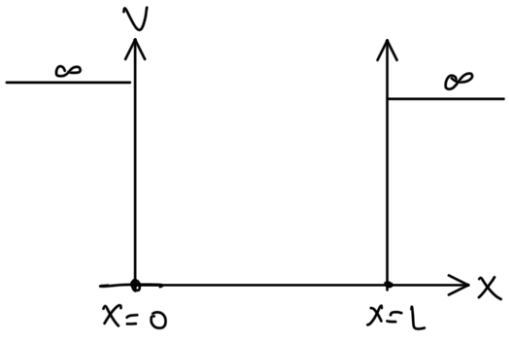
{ For a free electron, all possible energies were possible }

→ not normalizable ψ (as $E \to \infty$ so we can't talk about a definite energy

Particle in an Infinite potential well (particle in a box)

TDSWE:

$$\left(\frac{-\hbar^2}{2m}\frac{d^2}{dx^2} + V(x,t)\right)\psi(x,t) = i\hbar\frac{d}{dt}\psi(x,t)$$



$$V = \begin{cases} 0 & x \in (0, L) \\ \infty & \text{else} \end{cases}$$

An electron can't stay in a potential, hence $\psi = 0$ for $x \leq 0$ and $x \geq L$

$$\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \psi(x)V(x) = E\psi(x) \rightarrow \text{TISWE} \qquad \phi(t) = e^{-Et/\hbar}$$

$$\Rightarrow \left(\frac{-\hbar^2}{2m}\right)\frac{d^2\psi}{dx^2} = E\psi \Rightarrow \boxed{\frac{2mE}{\hbar^2} = \alpha^2}$$

$$\Rightarrow \alpha^2\psi = -\frac{d^2\psi}{dx^2} \Rightarrow \psi(x) = a_1 \sin\alpha x + a_2 \cos\alpha x$$

$\psi \rightarrow$ continuous, single valued

at L: $\psi(L) = 0 = a_1 \sin\alpha L + a_2 \cos\alpha L$

at 0: $\psi(0) = 0 = a_2 \Rightarrow \alpha L = n\pi \; ; \; n \in N$

$$\Rightarrow \psi(x) = A\sin\alpha x$$

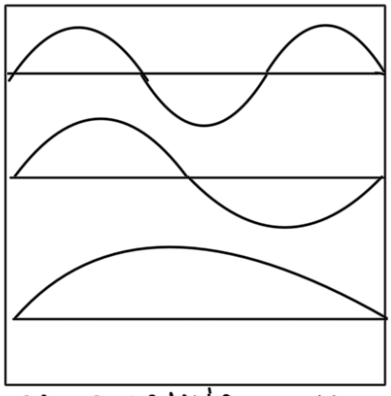
$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2} \Rightarrow \boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}}$$ Energy is

$\frac{\hbar^2}{}$ $\frac{}{L^2}$ $\boxed{\frac{}{2mL^2}}$ quantized

"Confinement (finite L) leads to quantization"

$\psi(x) = A \sin \frac{n\pi x}{L} \Rightarrow \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ (x ↑, n ↓)

$\int_0^L |\psi(x)|^2 dx = 1 \Rightarrow A^2 \times L/2 = 1 \rightarrow A = \sqrt{2/L}$



asymmetric well

→ ground state is not zero
→ Energy is discretized

momentum $\langle p \rangle = \int_0^L \left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}\right)\left(-i\hbar \frac{d}{dx}\right)\left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}\right)$

$= \int_0^L -i\hbar \frac{2}{L} \cos\left(\frac{n\pi x}{L}\right)\left(\frac{n\pi}{L}\right) \cdot \sin\left(\frac{n\pi x}{L}\right)$

$= \int_0^L \left(\frac{-i\hbar n}{L^2}\right) \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right)$

$= 0$

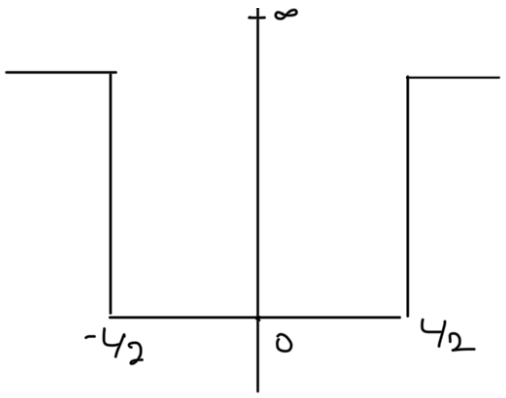
But $\langle p^2 \rangle = 2mE$

particle has to be blw 0 and L $\Rightarrow \Delta x = L$ (max)

$\Rightarrow \Delta \bar{p} = \frac{\hbar}{L} \Rightarrow E = \frac{\hbar^2}{2mL^2}$ → zero point energy

Symmetric well: (↑ ↓)

$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L} + \frac{n\pi}{2}\right)$



$$\psi_n(x) = \begin{cases} \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) & n \to \text{even} \\ \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right) & n \to \text{odd} \end{cases}$$

{particle in a box has KE; 0 PE}

properties of the ψ

① $\int_0^L \psi_i(x)\, \psi_j^*(x)\, dx = \delta_{ij} \Rightarrow$ orthonormality

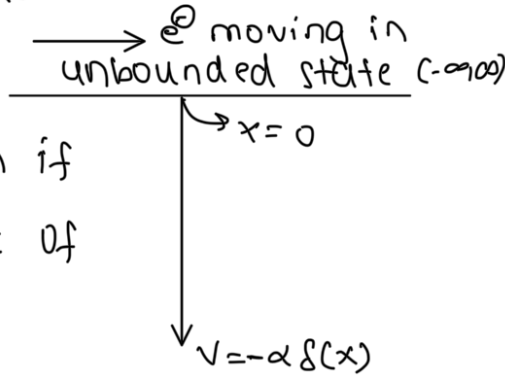
② Completeness: $\phi(x) = \sum_{n=1}^{\infty} c_n \psi_n(x)$ → fourier series

dirichlet theorm ← ↓ any wave-function $c_n \in \mathbb{F}$

$$c_n = \int_{-\infty}^{\infty} \phi(x)\, \psi_n^*(x)\, dx \qquad p = |c_n|^2$$

Dirac Delta function well:

$V(x) = \alpha\delta(x) \quad \alpha > 0, E < 0$ → (no reflection, scattering)



Some scattering happens even if energy of $e^{\ominus}$ is higher than that of the well

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \alpha\delta(x)\psi(x) = E\psi(x)$$

$x < 0 \Rightarrow -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x) \qquad K^2 = \frac{-2mE}{\hbar^2} \Rightarrow \frac{d^2\psi}{dx^2} = K^2\psi$

$x > 0$

$$\Rightarrow \psi(x) = Ae^{-Kx} + Be^{Kx}$$

$x<0 \Rightarrow A=0$

$x>0 \Rightarrow B=0$

$\Rightarrow$ $\psi(x)$ should be of the form

$$\Rightarrow \boxed{\psi(x) = Ae^{-k|x|}}$$

$$\frac{-\hbar^2}{2m}\int_{-\epsilon}^{\epsilon}\frac{d^2\psi(x)}{dx^2}dx + \int_{-\epsilon}^{\epsilon} V(x)\psi(x) = E\int_{-\epsilon}^{\epsilon}\psi(x)dx$$

$$\frac{\hbar^2}{2m}\left(D\frac{d\psi}{dx}\right) = \int_{-\epsilon}^{\epsilon} V(x)\psi(x)dx \Rightarrow \frac{\Delta d\psi}{dx} = -\frac{2m}{\hbar^2}\alpha \cdot A$$

$$\Rightarrow -2Ake^{-k\epsilon} = -\frac{2m\alpha A}{\hbar^2} \Rightarrow k = \frac{m\alpha}{\hbar^2}$$

$$\int_{-\infty}^{\infty}|\psi(x)|^2dx = 1 \Rightarrow \int_{-\infty}^{0} A^2e^{+kx}\cdot e^{kx}dx + \int_{0}^{\infty} A^2e^{-kx}e^{-kx} = \frac{A^2}{k}$$

$$\Rightarrow A = \sqrt{k} \Rightarrow A = \sqrt{m\alpha}/\hbar$$

$$k = \frac{m\alpha}{\hbar^2} = \frac{\sqrt{-2mE}}{\hbar} \Rightarrow E = -\frac{m\alpha^2}{2\hbar^2}$$ (only 1 energy eigenvalue)

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(r,t)\right]\psi(r,t) = i\hbar\frac{\partial}{\partial t}\psi(r,t)$$

For electron in 3D box:

$$\psi(r) = \psi(x)\psi(y)\psi(z) = \frac{\sqrt{8}}{\sqrt{L_xL_yL_z}}\sin\left(\frac{n\pi x}{L_x}\right)\sin\left(\frac{n\pi y}{L_y}\right)\sin\left(\frac{n\pi z}{L_z}\right)$$

Probability current density:

$$\frac{dS}{dt} = -\nabla J_p$$ (particle conservation)

$\psi(r,t)$: $\Rightarrow$ particle density $|\psi|^2$

$$\hat{H}\psi = E\hat{\psi}$$

$$\frac{d\psi}{dt} = \frac{1}{i\hbar}\hat{H}\psi \qquad \text{take conjugate: } \frac{d\psi^*}{dt} = -\frac{1}{i\hbar}\hat{H}^*\psi^*$$

$$\frac{dS}{dt} = \frac{d}{dt}(\psi^*\psi) = \psi^*\frac{d\psi}{dt} + \psi\frac{d\psi^*}{dt} = \psi^*\frac{1}{i\hbar}\hat{H}\psi - \psi\frac{1}{i\hbar}\hat{H}^*\psi^*$$

$$= \frac{1}{i\hbar}\left[\psi^*\hat{H}\psi - \psi\hat{H}^*\psi^*\right] \qquad \hat{H} = -\frac{\hbar^2\nabla^2}{2m} + V$$

if $V \to$ real & time independent

$$\Rightarrow \hat{H} = \hat{H}^*$$

$$\frac{dS}{dt} = \frac{1}{i\hbar}\left[\psi^*\left(\frac{-\hbar^2\nabla^2\psi}{2m} + V\psi\right) - \psi\left(\frac{-\hbar^2}{2m}\nabla^2\psi^* + V\psi^*\right)\right]$$

$$= \frac{1}{i\hbar}\left(\frac{-\hbar^2}{2m}\right)\left[\psi\nabla^2\psi^* - \psi^*\nabla^2\psi\right]$$

$$= \frac{i\hbar}{2m}\nabla\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right]$$

$$\Rightarrow \boxed{J_p = \frac{-i\hbar}{2m}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right]}$$

$$\psi(r,t) = \psi(r)\,e^{-iEt/\hbar}$$

$$J_p = \frac{-i\hbar}{2m}\left[\psi^*(r)\nabla\psi(r) - \psi(r)\nabla\psi^*(r)\right]$$

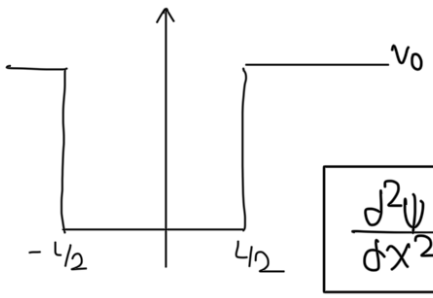
(time dependent terms cancel out)

$J_p \to$ time independent

→ Particle current density is stationery.

→ There is no radiation.

Finite Potential well



$E > V_0 \rightarrow$ continuum states

$E < V_0 \rightarrow$ bound states

$$\frac{d^2\psi}{dx^2} = k^2\psi$$

$E < V_0$: $$k = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

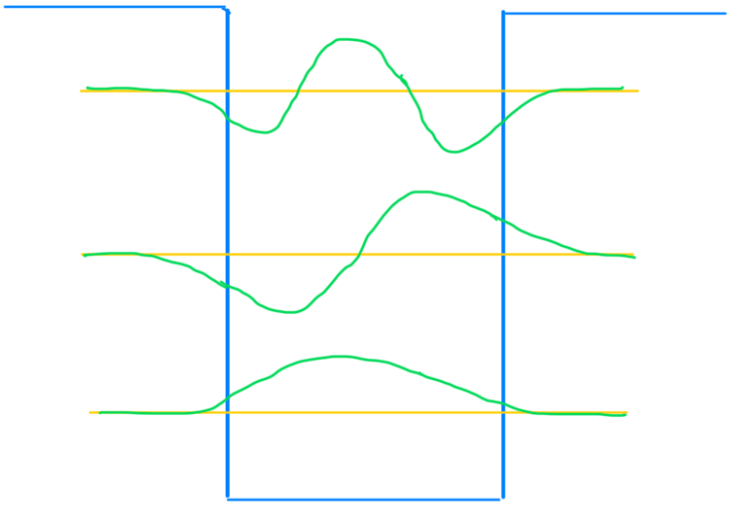
$\psi \rightarrow 0$ as $x \rightarrow \infty$

$\psi(z) = Ge^{kz}$; $x < -L/2$

$\psi(z) = A\sin kz + B\cos kz$; $x \in (-\frac{L}{2}, \frac{L}{2})$

$\psi(z) = Fe^{-kz}$; $x > L/2$

$\psi \rightarrow$ continuous, $\frac{d\psi}{dx} \rightarrow$ continuous



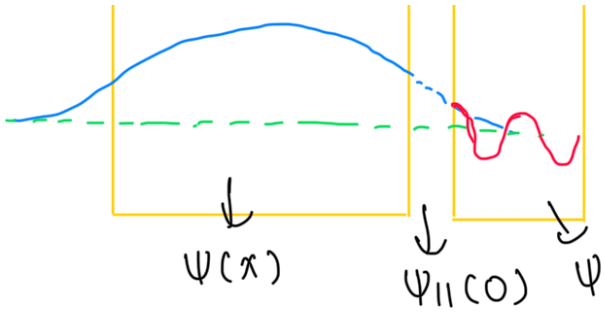
There's a finite probability that the ψ exists outside the finite-potential well

→ Eigenvalues are spaced uniformly for a parabolic potential well

→ wave functs are asymmetric for an asymmetric triangular potential well

Tunneling through a potential barrier:





$$\Rightarrow T = \frac{\ddot{\psi}_{11}(LB) \cdot \psi_{11}(LB)}{\psi_{11}^{*}(0) \cdot \psi_{11}(0)}$$

How Ohmic Contacts are made in ICs

Ohmic conductors → made by heavily doping surface region immediately beneath the contact. The barrier is small to enable tunneling

Tunneling is sometimes helpful and sometimes detrimental

